$\boxed{T9}$ 2 потока по 300 человек

| | I | II | |
|---|---|---|---|
| 2 | 33 | 39 | 72 |
| 3 | 43 | 35 | 78 |
| 4 | 80 | 72 | 152 |
| 5 | 144 | 154 | 298 |
| | 300 | 300 | |

$H_0$ : Потоки однородные

$H_1$ : $\overline{H_0}$

$\alpha = 0{,}05$

Для I: $\tilde{\Delta}_1 = \dfrac{\left(33 - 300 \cdot \frac{72}{600}\right)^2}{300 \cdot \frac{72}{600}} + \dfrac{\left(43 - 300 \cdot \frac{78}{600}\right)^2}{300 \cdot \frac{78}{600}} +$

$+ \dfrac{\left(80 - 300 \cdot \frac{152}{600}\right)^2}{300 \cdot \frac{152}{600}} + \dfrac{\left(144 - 300 \cdot \frac{298}{600}\right)^2}{300 \cdot \frac{298}{600}} = 1{,}039$

Для II: $\tilde{\Delta}_2 = \dfrac{\left(39 - 300 \cdot \frac{72}{600}\right)^2}{300 \cdot \frac{72}{600}} + \ldots + \dfrac{\left(154 - 300 \cdot \frac{298}{600}\right)^2}{300 \cdot \frac{298}{600}} =$

$= \cancel{1{,}039}\ 1{,}039$

$\Delta = \tilde{\Delta}_1 + \tilde{\Delta}_2 = 2{,}073$

$\Delta \sim \chi^2((2-1)(4-1)) = \chi^2(3)$

$p\text{-value} = \int_{2{,}073}^{+\infty} q(t)\,dt = 0{,}55 > \alpha = 0{,}05 \Rightarrow$

$\Rightarrow$ Нет оснований отвергать $H_0$